University of Florida, Department of Mathematics

Topology First-Year Exam, August 2025, Part 2

For the first five problems, show all your work and support all statements.

1. Suppose that the continuous maps $f, g : X \to Y$ are homotopic and that the continuous maps $h, k : Y \to Z$ are homotopic. Show that $h \circ f$ is homotopic to $k \circ g$.

2. Show that the fundamental group of the circle is isomorphic to the additive group of integers. That is, $\pi_1(S^1, 1) \cong \mathbb{Z}$.

3. State and prove the Brouwer fixed point theorem for the unit disk, D^2.

4. This problem has two parts.

(a) Define the Hawaiian earring space, H.

(b) Show the images of the generators of the fundamental groups of the circles do not generate the fundamental group of H.

5. This problem has two parts.

(a) Define surface.

(b) Compute the fundamental group of the connected sum of a torus and a second torus $T^2 \# T^2$.

Answer the following with complete definitions or statements or short proofs.

6. State the Urysohn Lemma.

7. Let E and B be topological spaces. Define what it means for a function $p : E \to B$ to be a covering map.

8. If $j : A \hookrightarrow X$ is the inclusion of a retract, then show that the induced map on fundamental groups is injective.

9. Let α be a path in X from x_0 to x_1. Define the map $\hat{\alpha} : \pi_1(X, x_0) \to \pi_1(X, x_1)$ and show that $\hat{\alpha}$ is a group isomorphism.

10. State the Seifert–van Kampen Theorem.

11. Let S^n be the n-sphere. Prove that $\pi_1(S^n)$ is trivial for $n \geq 2$.

12. State the Borsuk–Ulam Theorem for S^2.

13. State the Tychonoff theorem.

14. Describe a space X with $\pi(X) \cong \mathbb{Z}/3\mathbb{Z}$.

15. Define Stone–Čech compactification.